Practice 12. Binary Search

[CSE2010] Data Structures

Department of Data Science

Overview

- Implement **binary search**
  - Given a sequence A of integers and an integer x, find the position of x in A: O(logn) time

- Implement a function that checks if a given integer is a **perfect square**.
  - **Perfect square** is an integer that is the square of an integer; that is, a perfect square is the product of some integer with itself.
  - **Example**.
    9 is a perfect square since $3 \times 3 = 9$, but 8 is not a perfect square since it cannot be the product of some integer with itself.
  - **Constraint.**
    Use binary search to solve this problem.

Input and Output

- **B<space>[number_of_integers]<space>[x] [int]<space>[int], …, <space>[int]**
  - First, write “B<space>[number_of_integers]<space>[x]” into output file.
  - In the next line, if x does not exist in the sequence A, write “N”; otherwise, write the index of x.
- **P<space>[int]**
  - If the given integer is a perfect square, write “T” into output file.
  - Otherwise, write “F” into output file.

- Input File & Output File

```
B 5 7
2 4 5 7 9
B 5 1
2 4 5 7 9
B 1 3
3
P 121
P 110
P 1
```

```
B 5 7
3
B 5 1
N
B 1 3
0
T
F
T
```

Running Example

- Start from the skeleton code provided.

```
[hjkim@localhost search]$ make
g++ -std=c++11 -o practice10 search.cpp
[hjkim@localhost search]$ ./practice10 input1.txt output1.txt
[hjkim@localhost search]$ cat input1.txt
B 5 7
2 4 5 7 9
B 5 1
2 4 5 7 9
B 1 3
3
P 121
P 110
P 1
[hjkim@localhost search]$ cat output1.txt
B 5 7
3
B 5 1
N
B 1 3
0
T
F
T
[hjkim@localhost search]$
```

Submission Guideline

- Submission: **source code, an input file, an output file**
  - Where: Assignments in LMS
  - Deadline: **23:59, May. 26th (Sunday)**